

Introduction

Data → Information → Knowledge

Data:

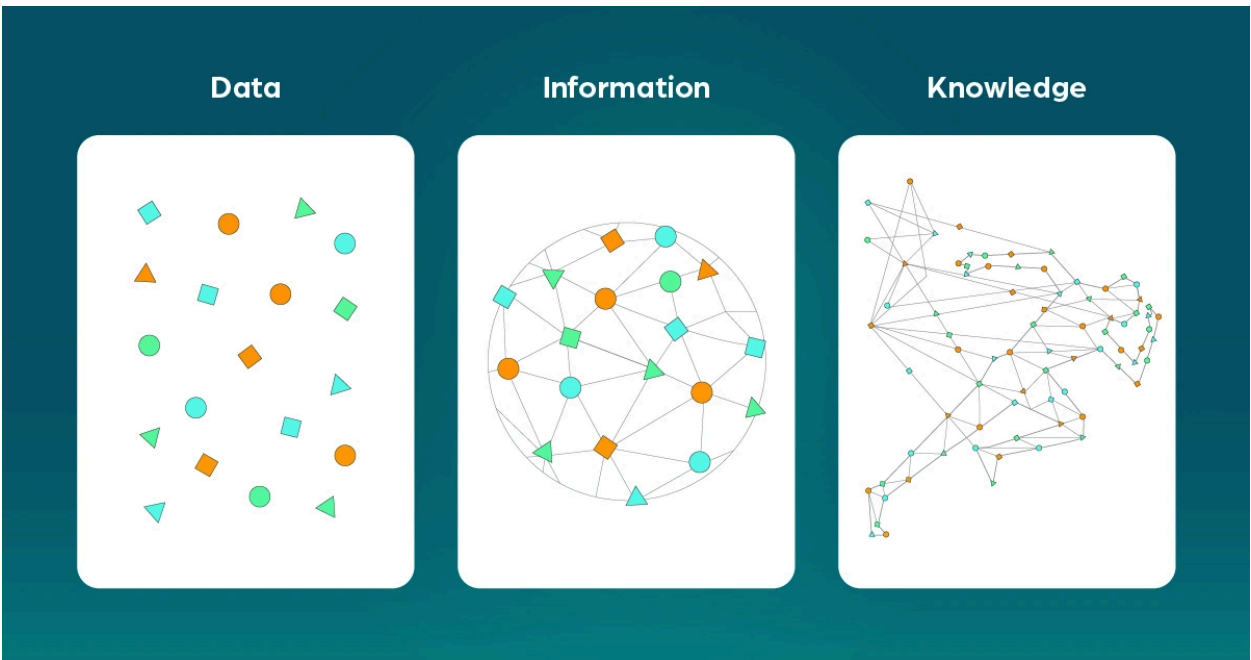
Raw facts with no meaning yet.
(Example: numbers, text, logs)

Information:

Data after processing and organizing it.
It becomes meaningful and useful.

Knowledge:

Deeper understanding extracted from information.
It represents **hidden patterns and insights**.

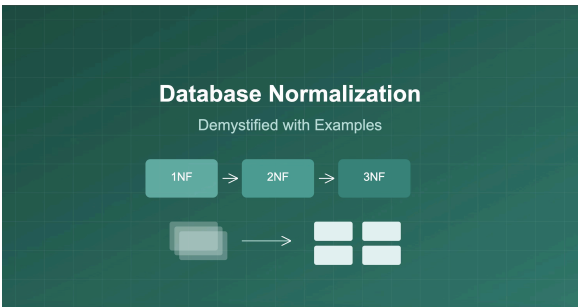


- **Knowledge = hidden patterns**
When you analyze information and discover patterns, you create knowledge.
- When you **share knowledge**, it becomes **information for others** (because they receive it as something structured and understandable).
- You can **process information multiple times**, and it can still remain **information** unless you discover new insights (knowledge).

What is Normalization?

Normalization is the process of organizing a database to reduce redundancy and improve data integrity.

<https://database.guide/what-is-normalization/>



Problems of Normalization in OLAP / DWH

1. Too many joins

- Queries need to combine many tables → **slower performance** for big datasets.

2. Complex queries

- Harder to write and maintain SQL queries.

3. Not ideal for reporting / analytics

- Analytical queries prefer **fewer tables**, so fewer joins = faster scans.